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TY - BCA Project

Event Management System

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Mini Project Report on

“Event Management System”

Submitted for the partial fulfillment of

Bachelor of Computer Application (BCA)

Project By

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Academic Year: 2025-26

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CERTIFICATE

This is to certify that the project work entitled **"Event Management System"** has been successfully completed for partial fulfillment of the Degree Course of **"Bachelor of Computer Application (BCA)**" of Kavayitri Bahinabai Chaudhari North Maharashtra University, Jalgaon. It has been carried out under my guidance.

It is the original work of **Mr. Lonagare Rohit Raju** who worked hard and sincerely completed the project work. I am fully satisfied with her performance.

Event Management System **Dr.Seema V. Rane**
Guide Name **Head of Department**

Examined By

Internal Examiner **External Examinerternal**

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DECLARATION

I hereby declare that the project work entitled "**Event Management System**" submitted to MTES's **Smt. G. G. Khadse College, Muktainagar**, is a record of an original work done by me with under guidance of **Dr. Ranjana S Zinjore**.

I further declare that this work has not been submitted in partly or fully to another university or college for the award of any other degree. Material obtained from other sources has been daily acknowledgement in the Field Work.

Name : Mr. Lonagare Rohit Raju

Date :   /   / 2026

Place : Muktainagar

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ABSTRACT

The College Event Management System is a web-based application developed to simplify and automate the management of events within educational institutions. Colleges frequently organize various academic and non-academic activities such as seminars, workshops, cultural programs, and sports events. Traditional event management methods often rely on manual processes like paper registrations and scattered communication, which can lead to inefficiencies, data loss, and lack of proper record management.

The proposed system provides a centralized digital platform where organizers can create and manage events while users can view and register for them online. The system follows a client–server architecture and integrates frontend, backend, and database components to ensure efficient data handling and real-time interaction. The frontend is developed to provide an intuitive interface for users, while the backend handles business logic and communication with the database for storing event and registration information.

The system includes key functionalities such as user authentication, event creation, event listing, and online registration. Organizers can monitor participant registrations and manage event details, whereas users can easily browse available events and participate by submitting their information through the system. The use of a structured database ensures data consistency, integrity, and easy retrieval of information.

Overall, the College Event Management System improves efficiency, reduces manual workload, and provides a more organized approach to managing college events. The project also demonstrates the practical implementation of concepts from software engineering, database management systems, and web development, making it both academically and practically valuable.

CHAPTER 1 : INTRODUCTION

1.1 Overview

In the present digital era, the management of academic and non-academic activities has undergone a significant transformation due to rapid advancements in information technology. Educational institutions such as colleges and universities frequently organize various events including seminars, workshops, cultural programs, technical fests, sports events, and guest lectures. Traditionally, the management of these events has been carried out using manual methods or semi-digital tools like paper forms, notice boards, spreadsheets, and informal communication platforms. Although these methods served their purpose in the past, they are no longer sufficient to handle the increasing scale, complexity, and data requirements of modern institutions.

The traditional approach to event management often involves manual registration, physical record maintenance, and verbal communication. These methods are time-consuming, error-prone, and lack transparency. For example, student registrations are usually recorded on paper or basic spreadsheets, which increases the chances of data loss, duplication, and inconsistency. Additionally, retrieving historical event data or tracking participant information becomes extremely difficult when records are not maintained in a structured digital format. Such limitations highlight the need for a centralized and automated event management solution.

With the widespread adoption of web technologies, web-based management systems have emerged as an effective solution to overcome these challenges. A web-based College Event Management System provides a centralized platform where events can be created, managed, and monitored efficiently. It allows different stakeholders—such as organizers and users—to interact with the system based on their roles and responsibilities. This role-based access ensures that the system remains secure, organized, and scalable.

From a theoretical perspective, a College Event Management System can be viewed as an application of concepts from software engineering, database management systems, and web application architecture. The system integrates front-end technologies for user interaction, back-end technologies for business logic processing, and databases for persistent data storage. Such integration ensures data consistency, reliability, and real-time availability of information.

In modern academic environments, students expect quick access to information and seamless digital services. A centralized event management platform fulfills this expectation by enabling users to view available events, register online, and track their participation history. Similarly, organizers benefit from tools that allow them to create events, manage registrations, and analyze participation data. Thus, the overall event management process becomes more efficient, transparent, and user-friendly.

Therefore, the College Event Management System represents a shift from traditional manual practices to a structured, automated, and technology-driven approach. It aligns with current trends in digital transformation within educational institutions and reflects the growing importance of information systems in academic administration.

1.2 Purpose of the System

The primary purpose of the College Event Management System is to provide a reliable and efficient platform for managing college-level events in a systematic manner. The system aims to replace conventional manual processes with a fully digital solution that simplifies event creation, registration, and monitoring. By doing so, it reduces administrative workload and enhances operational efficiency.

One of the key purposes of the system is to centralize event-related information. In traditional setups, event details are scattered across notice boards, emails, or messaging applications, making it difficult for students to stay informed. The proposed system stores all event-related data in a single database, ensuring consistency and easy accessibility. This centralized approach allows users to view accurate and up-to-date information at any time.

Another important purpose is to improve data accuracy and integrity. Manual data entry often leads to errors such as duplicate records, missing information, or incorrect entries. A structured digital system enforces validation rules and standardized data formats, thereby minimizing such errors. This ensures that the data stored in the system remains reliable and meaningful for future reference.

The system also aims to support role-based functionality. Different users interact with the system in different ways. Organizers are responsible for creating and managing events, while users primarily register and participate in events. By clearly defining these roles, the system ensures that users can only access features relevant to their responsibilities. This not only improves usability but also enhances security.

From an academic perspective, the purpose of the system extends beyond mere automation. It serves as a practical implementation of theoretical concepts learned in subjects such as web development, database systems, and software engineering. The project demonstrates how abstract concepts like client-server architecture, CRUD operations, and data modeling are applied in real-world applications.

In summary, the purpose of the College Event Management System is to create a structured, efficient, and secure environment for managing events. It bridges the gap between traditional administrative methods and modern digital solutions, thereby improving the overall quality of event management within educational institutions.

1.3 Objectives of the System

The objectives of the College Event Management System can be broadly categorized into functional, technical, and academic objectives. Each category addresses a specific aspect of system development and usage.

1.3.1 Functional Objectives

The functional objectives focus on what the system is expected to achieve from a user and organizer perspective. One of the primary functional objectives is to enable organizers to create and manage events efficiently. This includes defining event details such as title, date, venue, and description, as well as maintaining control over event lifecycle activities.

Another important functional objective is to provide users with an easy-to-use interface for viewing and registering for events. Users should be able to browse available events, understand event details, and register without unnecessary complexity. This improves participation rates and ensures smooth coordination between organizers and participants.

The system also aims to maintain accurate registration records. By storing participant details in a structured format, the system enables organizers to view registered users for each event. This supports better planning, communication, and post-event analysis.

1.3.2 Technical Objectives

From a technical standpoint, the system aims to implement a modular and scalable architecture. The use of a client-server model separates the presentation layer from the business logic and database layer. This separation improves maintainability and allows future enhancements without major structural changes.

Another technical objective is to ensure data persistence and integrity through the use of a database management system. Structured collections and relationships ensure that event and registration data is stored efficiently and can be retrieved quickly when needed.

Security is also a critical technical objective. The system aims to restrict access based on user roles and prevent unauthorized operations. This is achieved through authentication mechanisms and controlled access to system functionalities.

1.3.3 Academic Objectives

From an academic perspective, the project aims to provide hands-on experience in designing and developing a real-world application. It reinforces theoretical concepts taught in coursework by applying them in a practical context. Students gain exposure to system analysis, design, implementation, testing, and documentation.

The project also aims to improve problem-solving and analytical skills by addressing real-life challenges in event management. By designing a complete system from scratch, students develop a deeper understanding of software development life cycle (SDLC) principles.

1.4 Future Scope

The College Event Management System has significant potential for future enhancements. As technology continues to evolve, the system can be extended to include additional features and integrations.

One possible enhancement is the integration of online payment gateways. This would allow users to register for paid events directly through the system, making it suitable for large-scale events and workshops. Another potential feature is automated certificate generation, which would simplify post-event processes and improve user satisfaction.

The system can also be extended to support mobile platforms. A dedicated mobile application would increase accessibility and allow users to receive real-time notifications about upcoming events. Additionally, advanced analytics and reporting tools could be incorporated to help organizers analyze participation trends and improve future event planning.

From a deployment perspective, the system can be migrated to cloud infrastructure to improve scalability, availability, and performance. Cloud deployment would also enable institutions to manage multiple campuses or departments through a single unified platform.

In conclusion, the future scope of the College Event Management System is vast. With continuous improvements and technological advancements, the system can evolve into a comprehensive event management solution suitable for diverse institutional needs.

CHAPTER 2 : SYSTEM ANALYSIS

2.1 Introduction to System Analysis

System analysis is one of the most critical phases in the software development life cycle (SDLC). It focuses on understanding the existing system, identifying its limitations, and defining the requirements of a new and improved system. The primary objective of system analysis is to study the current operational environment and determine how a proposed system can solve existing problems more efficiently.

In the context of a College Event Management System, system analysis plays a vital role because event management involves multiple stakeholders, data flows, and operational constraints. Without a proper analysis, the developed system may fail to meet user expectations or institutional requirements. Therefore, a detailed system analysis ensures that the final system is reliable, scalable, and aligned with real-world needs.

This chapter discusses the analysis of the existing event management practices, highlights their drawbacks, and presents the proposed system along with its requirements. The analysis is based on theoretical concepts of information systems, data management, and user interaction models.

2.2 Existing System

2.2.1 Overview of the Existing System

The existing system of event management in most colleges is largely manual or semi-digital in nature. Events are typically announced through notice boards, classroom announcements, or informal digital platforms such as messaging applications. Registration for events is often done using paper forms or basic spreadsheets maintained by organizers. While these methods are simple, they are not suitable for handling large volumes of data or ensuring long-term data consistency.

In many cases, student participation details are collected manually and stored in physical files. This makes data retrieval slow and inefficient. Additionally, there is no centralized repository for event-related information, leading to data fragmentation and redundancy.

2.2.2 Limitations of the Existing System

One of the major limitations of the existing system is **lack of automation**. Manual processes require significant human effort and time. Organizers must repeatedly perform tasks such as collecting forms, verifying entries, and updating records, which increases the chances of errors.

Another significant drawback is **poor data security**. Physical records and unprotected digital files are vulnerable to unauthorized access, loss, or damage. There is no mechanism to restrict access based on user roles, which can lead to misuse of sensitive information.

The existing system also suffers from **inefficient communication**. Students may miss important event announcements due to limited visibility or delayed communication. Similarly, organizers find it difficult to track the number of participants or manage last-minute changes.

From a theoretical standpoint, the existing system fails to satisfy key principles of information systems such as data integrity, consistency, and accessibility. The absence of structured databases and standardized workflows limits the system's effectiveness and scalability.

2.3 Proposed System

2.3.1 Overview of the Proposed System

The proposed College Event Management System is a web-based application designed to overcome the limitations of the existing system. It provides a centralized platform where events can be created, managed, and accessed digitally. The system is based on a client-server architecture, ensuring separation of concerns between user interface, business logic, and data storage.

In the proposed system, all event-related data is stored in a centralized database, which ensures consistency and easy retrieval. Users interact with the system through a web interface, making it accessible from anywhere with an internet connection.

2.3.2 Features of the Proposed System

One of the key features of the proposed system is **role-based access control**. Different users interact with the system based on predefined roles. This ensures that each user has access only to the functionalities relevant to their responsibilities.

Another important feature is **online event registration**. Users can view available events and register digitally, eliminating the need for physical forms. This improves efficiency and reduces administrative workload.

The system also provides **real-time data updates**, allowing organizers to monitor registrations and manage events effectively. This ensures better planning and decision-making.

From a theoretical perspective, the proposed system adheres to principles of modularity, scalability, and maintainability. It is designed to support future enhancements without major structural changes.

2.3.3 Advantages of the Proposed System

The proposed system offers several advantages over the existing system. It significantly reduces manual effort and minimizes errors through automation. Centralized data storage improves data security and integrity.

The system also enhances user experience by providing a simple and intuitive interface. Faster data retrieval and improved communication lead to higher user satisfaction and increased participation in events.

2.4 System Requirements

System requirements define what the system must do and how it should perform. These requirements are categorized into functional and non-functional requirements.

2.6.1 Functional Requirements

Functional requirements describe the specific operations that the system must support. These requirements are derived from user needs and operational workflows.

The system should allow organizers to create, update, and delete events. Each event should contain essential details such as title, date, venue, and description. The system should store this information in a structured format.

Users should be able to view a list of available events and register for events of interest. The system should capture user details during registration and store them securely in the database.

The system should also allow organizers to view registered participants for each event. This supports effective event coordination and planning.

2.6.2 Non-Functional Requirements

Non-functional requirements define the quality attributes of the system. These include performance, security, usability, and scalability.

The system should be responsive and capable of handling multiple users simultaneously without performance degradation. Efficient database queries and optimized data handling are essential to achieve this goal.

Security is another critical requirement. The system should protect user data from unauthorized access and ensure that sensitive information is stored securely.

Usability is equally important. The user interface should be simple, intuitive, and easy to navigate, even for users with limited technical knowledge.

Scalability ensures that the system can accommodate future growth, such as an increase in the number of users or events. The proposed architecture supports scalability through modular design and centralized data management.

CHAPTER 3 : SYSTEM DESIGN

3.1 Introduction to System Design

System design is the phase that transforms the requirements identified during system analysis into a detailed blueprint for implementation. While system analysis focuses on *what* the system should do, system design focuses on *how* the system will achieve those objectives. It defines the architecture, components, interfaces, data flow, and operational logic of the system.

In the College Event Management System, system design plays a crucial role because the application involves multiple user roles, dynamic data handling, and continuous interaction between frontend and backend components. A well-structured system design ensures that the application remains maintainable, scalable, and efficient throughout its lifecycle.

This chapter presents the architectural design, modular design, data design, and interface design of the proposed system. The design decisions are based on standard software engineering principles such as modularity, abstraction, cohesion, and low coupling.

3.2 System Architecture Design

3.2.1 Overview of System Architecture

The proposed College Event Management System follows a **client–server architecture**. In this architecture, the system is divided into three major layers:

1. Presentation Layer
2. Application (Business Logic) Layer
3. Data Layer

This layered architecture ensures separation of concerns, which improves maintainability and scalability.

The presentation layer handles user interaction through a web interface. The application layer processes requests, applies business rules, and coordinates data exchange. The data layer is responsible for persistent storage and retrieval of information.

3.2.2 Client–Server Architecture

In the client–server model, the client sends requests to the server, and the server processes those requests and returns appropriate responses. This model is widely used in modern web applications due to its flexibility and efficiency.

In this system:

- The **client** represents the web-based user interface accessed through a browser.
- The **server** manages application logic, authentication, and database communication.

This architecture allows centralized control over data and logic, while clients remain lightweight and focused on user interaction.

3.2.3 Advantages of the Chosen Architecture

The client–server architecture offers several advantages:

- Centralized data management ensures consistency and integrity.
- Security policies can be enforced at the server level.
- The system can support multiple users simultaneously.
- Future enhancements can be implemented without major changes to the client.

From a theoretical perspective, this architecture aligns with distributed system principles and supports scalability and fault tolerance.

3.3 Modular Design

3.3.1 Concept of Modular Design

Modular design involves breaking the system into smaller, independent units called modules. Each module performs a specific function and interacts with other modules through well-defined interfaces.

In software engineering, modularity improves readability, reusability, and ease of maintenance. Changes in one module have minimal impact on others, provided interfaces remain unchanged.

3.3.2 Modules in the System

The College Event Management System is divided into the following major modules:

1. Authentication Module
2. Organizer Module
3. User Module
4. Event Management Module

5. Registration Management Module

Each module is designed to handle a specific set of responsibilities.

3.3.3 Authentication Module

The authentication module manages user login and registration. It verifies user credentials and determines access rights based on predefined roles.

This module ensures that only authorized users can access the system. It also maintains user session integrity and prevents unauthorized access to restricted functionalities.

Theoretical concepts such as identity verification and access control are applied in this module to ensure system security.

3.3.4 Organizer Module

The organizer module allows authorized organizers to manage events. This includes creating events, viewing created events, deleting events, and monitoring registrations.

This module follows the principle of **high cohesion**, as all functionalities related to event organization are grouped together. It interacts with the event management module and registration module to retrieve and display relevant data.

3.3.5 User Module

The user module enables users to view available events and register for events of interest. It focuses on providing a simple and intuitive interface for participation.

This module emphasizes usability and responsiveness. It ensures that users can easily access event information without unnecessary complexity.

3.4 Data Design

3.6.1 Importance of Data Design

Data design defines how data is structured, stored, and accessed within the system. A well-designed data model ensures efficient storage, faster retrieval, and reduced redundancy.

In this system, data design is crucial because event details, user information, and registrations must be handled accurately and securely.

3.6.2 Database Design Approach

The system uses a **document-oriented database design**, which stores data in structured documents rather than rigid tables. This approach is suitable for applications with evolving data structures and flexible requirements.

Each major entity in the system is represented as a separate collection. Relationships between entities are maintained using unique identifiers.

3.6.3 Data Entities

The major data entities include:

- User Information
- Event Details
- Event Registrations

Each entity contains attributes relevant to its role in the system. Proper indexing ensures efficient query execution and performance optimization.

3.6.4 Data Integrity and Consistency

Data integrity ensures that stored data remains accurate and reliable. The system enforces integrity by validating inputs before storage and maintaining consistent data formats.

Consistency is maintained by centralizing data storage and avoiding duplicate records. This aligns with database normalization principles and information management theory.

3.5 Interface Design

3.6.1 User Interface Design Principles

User interface design focuses on creating an intuitive and visually appealing interaction environment. The system follows standard UI principles such as simplicity, consistency, and feedback.

Clear labels, logical navigation, and responsive layouts ensure a positive user experience.

3.6.2 Organizer Interface Design

The organizer interface is designed to support efficient event management. It provides clear sections for event creation, event listing, and registration viewing.

Visual separation of components improves readability and usability. Action buttons are placed strategically to reduce user effort.

3.6.3 User Interface Design

The user interface emphasizes simplicity and clarity. Users can easily browse events and register without navigating through complex menus.

Consistent styling and layout across pages ensure a uniform user experience.

3.6 Security Design

3.7.1 Security Considerations

Security design ensures protection of user data and system resources. The system incorporates access control mechanisms to restrict unauthorized actions.

Only authenticated users can access protected functionalities, and sensitive operations are restricted based on user roles.

3.7.2 Data Protection

User credentials and personal data are handled carefully to prevent misuse. Secure communication between client and server ensures data confidentiality.

These measures align with theoretical concepts of information security and data protection.

3.7 Summary of System Design

System design translates conceptual requirements into a structured blueprint for implementation. The proposed design ensures modularity, scalability, and maintainability.

By adopting a layered architecture and modular approach, the College Event Management System achieves flexibility and robustness. This design serves as a strong foundation for implementation and future enhancements.

CHAPTER 4 : SECURITY

4.1 Password Hashing using Bcrypt

In the proposed system, user passwords are not stored in plain text format. Instead, they are securely hashed using the bcrypt algorithm before storing them in the database. Bcrypt applies salting and multiple hashing rounds, which makes it computationally difficult for attackers to decode the original password. During login, the entered password is compared with the stored hashed password using bcrypt.compare(), ensuring secure authentication.

4.2 JWT-Based Authentication

The system uses JSON Web Token (JWT) for authentication. When a user logs in successfully, the server generates a token using jwt.sign(), which contains user information such as email and role. This token is sent to the client and stored locally. For every subsequent request, the token is sent in the request header and verified on the server side using jwt.verify(), ensuring that only authenticated users can access protected resources.

4.3 Role-Based Access Control (RBAC)

The application implements role-based authorization by assigning roles such as "organizer" and "user" to different users. Based on the role, access to certain routes and functionalities is restricted. For example, only organizers can manage events, while normal users can only view or register. This improves system security by limiting unauthorized access.

4.4 Authentication Middleware (Backend Security Layer)

A dedicated middleware is implemented in the backend to protect secure routes. This middleware extracts the JWT token from the request header, verifies it, and allows access only if the token is valid. If the token is missing or invalid, access is denied. This ensures that all protected APIs are accessed only by authenticated users.

4.5 Angular Auth Guard (Frontend Protection)

On the frontend, an auth guard is implemented to prevent unauthorized users from accessing restricted pages. Before navigating to protected routes like organizer or user dashboard, the guard checks whether a valid token exists and whether the user has the required role. If not, the user is redirected to the login page.

4.6 HTTP Interceptor for Secure Communication

An HTTP interceptor is used in the Angular application to automatically attach the JWT token to every outgoing HTTP request. This removes the need to manually include tokens in each request and ensures consistent and secure communication between frontend and backend.

4.7 Input Validation Mechanism

The system includes input validation both on the client side and server side. Email format validation and password length checks are implemented to ensure that only valid data is processed. This reduces the chances of invalid inputs and prevents common vulnerabilities such as malformed data submission.

4.8 Token-Based Session Management

Instead of using traditional session management, the system uses token-based sessions. Once a user logs in, the token acts as proof of authentication. This approach is stateless, scalable, and efficient, as the server does not need to store session data.

4.9 Error Handling and Security Feedback

The system provides appropriate error messages such as "Invalid email or password" without revealing sensitive internal details. This helps maintain security while guiding the user correctly.

CHAPTER 5 : SOFTWARE REQUIREMENTS SPECIFICATION (SRS)

5.1 Introduction

Software Requirements Specification (SRS) is one of the most important documents in the development of any software system. It provides a detailed and precise description of the system's expected behavior, functionalities, and quality attributes. From a theoretical perspective, SRS acts as a foundation upon which the entire system is designed, implemented, and tested. A well-written SRS ensures that all stakeholders have a common understanding of the system requirements and reduces ambiguity during development.

In the College Event Management System, the SRS document plays a crucial role because the system deals with multiple users, dynamic event data, and continuous interactions between different components. Event management requires accurate handling of information such as user credentials, event details, and registration records. Therefore, defining requirements clearly is essential to ensure data consistency, system reliability, and user satisfaction.

This chapter presents the Software Requirements Specification of the proposed system in a descriptive and analytical manner. The focus is on explaining *what the system must do* and *how well it should perform*, without discussing implementation details.

5.2 Purpose of the Software Requirements Specification

The primary purpose of preparing the SRS for the College Event Management System is to formally document the system requirements in a clear and understandable manner. This document serves as a reference for developers during implementation and for evaluators during project assessment. From an academic standpoint, it demonstrates the application of requirement engineering principles taught in software engineering courses.

Another important purpose of the SRS is to ensure correctness and completeness of system requirements. By documenting requirements in advance, potential misunderstandings between users and developers can be avoided. This is particularly important in systems like event management, where even minor errors in requirements may lead to incorrect handling of registrations or loss of important data.

The SRS also acts as a baseline for testing activities. Each requirement described in this document can be verified during system testing, ensuring that the developed system behaves as expected. Thus, the SRS contributes directly to the quality assurance of the software product.

5.3 Scope of the System

The scope of the College Event Management System defines the boundaries of the system and identifies the functionalities that are included. The system is intended to manage academic and non-academic events conducted within a college environment. It provides a centralized digital platform for handling event creation, user registration, and participant management.

The system focuses on automating activities that were traditionally handled manually or through unorganized digital tools. It covers functionalities such as user authentication, event listing, online registration, and viewing of registered participants. By limiting the scope to core event management activities, the system remains simple, efficient, and suitable for academic use.

At the same time, the system is designed with extensibility in mind. Although advanced features such as online payments or automated certificate generation are not included in the current scope, the system architecture supports future enhancements.

5.4 Overall Description of the System

5.4.1 Product Perspective

From a theoretical viewpoint, the College Event Management System can be considered a standalone web-based information system. It follows a client-server model, where users interact with the system through a web interface, and the server handles data processing and storage. The system does not depend on any existing legacy system and maintains its own database.

The separation between client and server ensures better organization of system components. The client focuses on user interaction and presentation, while the server manages business logic and data access. This separation aligns with standard software architecture principles and improves system maintainability.

5.4.2 Product Functions

The system provides a set of core functions that collectively support event management operations. One of the primary functions is user authentication, which ensures that only authorized users can access the system. Authentication is essential to protect sensitive information and maintain system security.

Another major function is event management, where authorized users can create and manage event details. This function ensures that all events are stored in a structured format and can be accessed easily by participants. Additionally, the system supports online event registration, allowing users to register without physical paperwork.

The system also provides functionality for viewing registration data. This allows organizers to analyze participation and manage events more effectively. All these functions work together to provide a complete and integrated event management solution.

5.4.3 User Classes and Characteristics

The system identifies two primary user classes based on their interaction with the system. The first user class consists of organizers, who are responsible for managing events. Organizers require access to functionalities such as event creation, deletion, and viewing of registered participants. They are assumed to have basic familiarity with web applications.

The second user class includes general users, who primarily interact with the system to view available events and register for participation. These users may not possess advanced technical skills; therefore, the system interface must be simple and intuitive.

By clearly defining user classes, the system ensures that access rights are controlled and system misuse is prevented.

5.4.4 Operating Environment

The College Event Management System operates in a web-based environment and is accessible through standard web browsers. The system requires an active internet connection for communication between the client and server components.

The server environment supports backend processing and database management. The system is designed to be platform-independent, allowing users to access it from different devices without compatibility issues.

5.5 Functional Requirements (Narrative Explanation)

Functional requirements describe the expected behavior of the system. In the proposed system, one of the most important functional requirements is the ability to register new users. The system must allow users to create an account by providing valid credentials, which are then stored securely in the database.

Another critical requirement is user authentication. The system must verify user credentials during login and grant access only to authenticated users. This ensures that unauthorized users cannot access system functionalities.

Event creation is another key functional requirement. The system must allow authorized users to define event details such as title, date, venue, and description. These details must be stored accurately and made available to participants.

The system must also support event viewing and registration. Users should be able to browse available events and register for them online. Registration details must be associated with the corresponding event to maintain data consistency.

Finally, the system must provide a mechanism for viewing registered participants. This supports effective event coordination and evaluation.

5.6 Non-Functional Requirements (Quality Attributes)

Non-functional requirements define how well the system performs its functions. Performance is a key requirement, as the system must handle multiple users simultaneously without significant delays. Efficient data processing and optimized database access contribute to acceptable system performance.

Security is another essential requirement. The system must protect user data from unauthorized access and ensure that sensitive information is handled securely. Authentication and access control mechanisms support this requirement.

Usability is also critical, especially for users with limited technical knowledge. The system must provide a simple and consistent interface that allows users to perform tasks easily.

Reliability and scalability are additional non-functional requirements. The system must function consistently over time and support future growth in terms of users and data volume.

5.7 Constraints, Assumptions, and Dependencies

The system operates under certain constraints, such as dependence on internet connectivity and limited hardware resources. Time constraints associated with academic project development also influence system design decisions.

The system assumes that users have basic access to web browsers and internet services. It also depends on proper server configuration and database availability for smooth operation.

CHAPTER 6 : METHODOLOGY

6.1 Introduction to Methodology

Methodology refers to the systematic and structured approach adopted for the development of a software system. It defines the logical framework through which a problem is analyzed, designed, implemented, and evaluated. From a theoretical perspective, methodology is not merely a sequence of steps but a disciplined process that ensures the development of a reliable, efficient, and maintainable system.

In the development of the College Event Management System, the methodology plays a crucial role in transforming conceptual requirements into a functional software product. Since the system handles sensitive information such as user credentials and event registrations, a well-defined methodology is essential to maintain data integrity, usability, and security. This chapter discusses the development methodology adopted for the system in a descriptive and analytical manner.

6.2 Selection of Development Methodology

The selection of an appropriate development methodology is a critical decision in software engineering. Different methodologies such as Waterfall, Agile, Spiral, and Incremental models are available, each with its own strengths and limitations. For academic projects like the College Event Management System, the methodology must balance theoretical clarity with practical feasibility.

A structured and phased approach was selected for the development of this system. This approach allows systematic progression from requirement analysis to system design, implementation, testing, and documentation. From a theoretical standpoint, such an approach ensures that each phase is completed and validated before moving to the next, thereby reducing errors and inconsistencies.

The chosen methodology emphasizes clarity of requirements, modular development, and continuous validation. This makes it suitable for systems that involve multiple modules and user interactions, such as event management platforms.

6.3 Requirement Analysis Phase

The requirement analysis phase forms the foundation of the entire development process. In this phase, the functional and non-functional needs of the system are identified and documented. For the College Event Management System, this phase involved studying existing event management practices and identifying their limitations.

From a theoretical perspective, requirement analysis aims to bridge the gap between user expectations and system capabilities. It focuses on understanding what the system should achieve rather than how it should be implemented. This phase ensures that the developed system addresses real-world problems effectively.

During requirement analysis, emphasis was placed on identifying core functionalities such as user authentication, event creation, and online registration. Non-functional aspects such as usability, security, and scalability were also considered to ensure overall system quality.

6.4 System Design Phase

The system design phase translates analyzed requirements into a structured architectural blueprint. This phase defines how different components of the system interact with each other and how data flows within the system. From a methodological viewpoint, system design acts as a bridge between conceptual requirements and actual implementation.

In the College Event Management System, the design phase focused on modularization and separation of concerns. Each major functionality was conceptualized as an independent module, allowing easier maintenance and future enhancements. Theoretical principles such as high cohesion and low coupling were applied to ensure system robustness.

Design considerations also included user interface structure, data organization, and access control mechanisms. These considerations ensure that the system remains user-friendly while maintaining operational efficiency.

6.5 Implementation Strategy

Implementation refers to the process of converting system design into an operational software system. From a methodological perspective, implementation is guided by predefined design decisions and coding standards. It focuses on accuracy, consistency, and adherence to system specifications.

In this project, the implementation strategy followed a modular approach, where individual components were developed and validated independently. This approach aligns with theoretical concepts of incremental development, where system functionality is built progressively.

Special attention was given to data handling and user interaction during implementation. Proper validation mechanisms ensure that only accurate and complete data is stored in the system. This strategy enhances system reliability and reduces the chances of runtime errors.

6.6 Testing Methodology

Testing is an integral part of the development methodology and ensures that the system functions as expected. From a theoretical standpoint, testing aims to identify defects, validate requirements, and ensure system stability.

The testing methodology adopted for the College Event Management System focuses on functional testing and integration testing. Each module is tested independently to verify its correctness. After individual testing, modules are integrated and tested as a complete system to ensure smooth interaction.

Testing also considers user scenarios to evaluate system behavior under real-world conditions. This helps in identifying usability issues and improving overall system performance.

6.7 Validation and Verification

Validation and verification are essential components of the methodology. Verification ensures that the system is built correctly according to specifications, while validation ensures that the right system has been built to meet user needs.

In this project, verification is achieved by comparing system behavior with documented requirements. Validation is achieved by evaluating system performance from a user perspective. This dual approach ensures both technical accuracy and practical usability.

6.8 Documentation Process

Documentation is a continuous activity throughout the development process. From a methodological viewpoint, proper documentation ensures knowledge transfer, maintainability, and academic evaluation.

For the College Event Management System, documentation includes requirement specifications, system design descriptions, and user guidance. This documentation not only supports development but also serves as evidence of systematic project execution.

CHAPTER 7 : SYSTEM MODULES

7.1 Introduction

A software system is usually divided into smaller functional units called modules. Modular design improves the maintainability, scalability, and clarity of the system by separating different functionalities into independent components. Each module performs a specific task and interacts with other modules through well-defined interfaces.

In the College Event Management System, the system is divided into several modules that collectively handle user authentication, event management, and event participation. These modules ensure that the system operates efficiently and provides a structured workflow for both organizers and users.

7.2 Authentication Module

The authentication module is responsible for user registration and login functionality. It verifies the credentials provided by users and grants access to the system based on their role.

Users can create accounts by providing their email and password. During login, the system validates these credentials and allows access only if the information is correct. This module ensures system security and prevents unauthorized access.

7.3 Organizer Module

The organizer module allows authorized organizers to manage events. Organizers have the ability to create new events, view existing events, and delete events if necessary.

This module provides organizers with tools to control event-related activities and monitor participation. It ensures that event information remains organized and easily manageable.

7.4 User Module

The user module enables users (students) to interact with the system. Users can browse available events, view event details, and register for events of their interest.

The module is designed with simplicity and usability in mind so that users can easily access event information and participate without technical difficulty.

7.5 Event Management Module

The event management module handles the creation and maintenance of event data. Organizers can add event details such as title, date, venue, and description.

This module ensures that event information is stored correctly and displayed to users in a structured manner.

7.6 Registration Management Module

The registration management module manages user participation in events. When a user registers for an event, their details are stored in the database and linked to the selected event.

Organizers can view the list of registered participants for each event, allowing them to manage attendance effectively.

CHAPTER 8 : SYSTEM DIAGRAMS

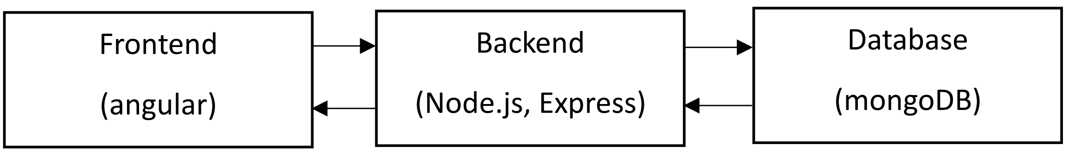
8.1 Introduction:

System diagrams are graphical representations used to illustrate the structure and flow of information within a system. These diagrams help in understanding how different components of the system interact with each other.

In the College Event Management System, several diagrams are used to explain the system architecture, data flow, and relationships between entities.

8.2 Architecture Diagram :

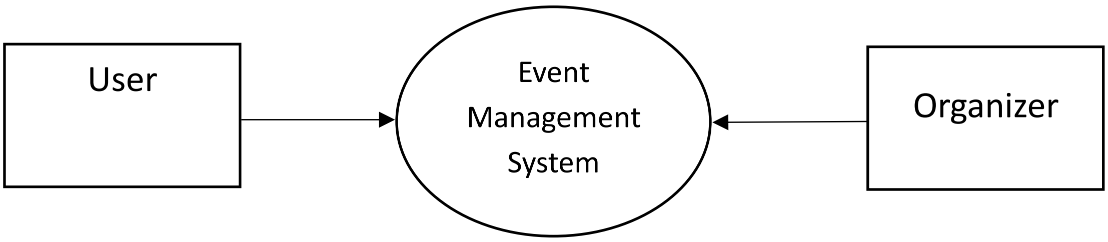
The architecture diagram shows the overall structure of the system. The system follows a three-tier architecture, which consists of:



8.3 DFD Level 0 :

The Level 0 DFD represents the overall interaction between external entities and the system. In this diagram, two main entities interact with the system.

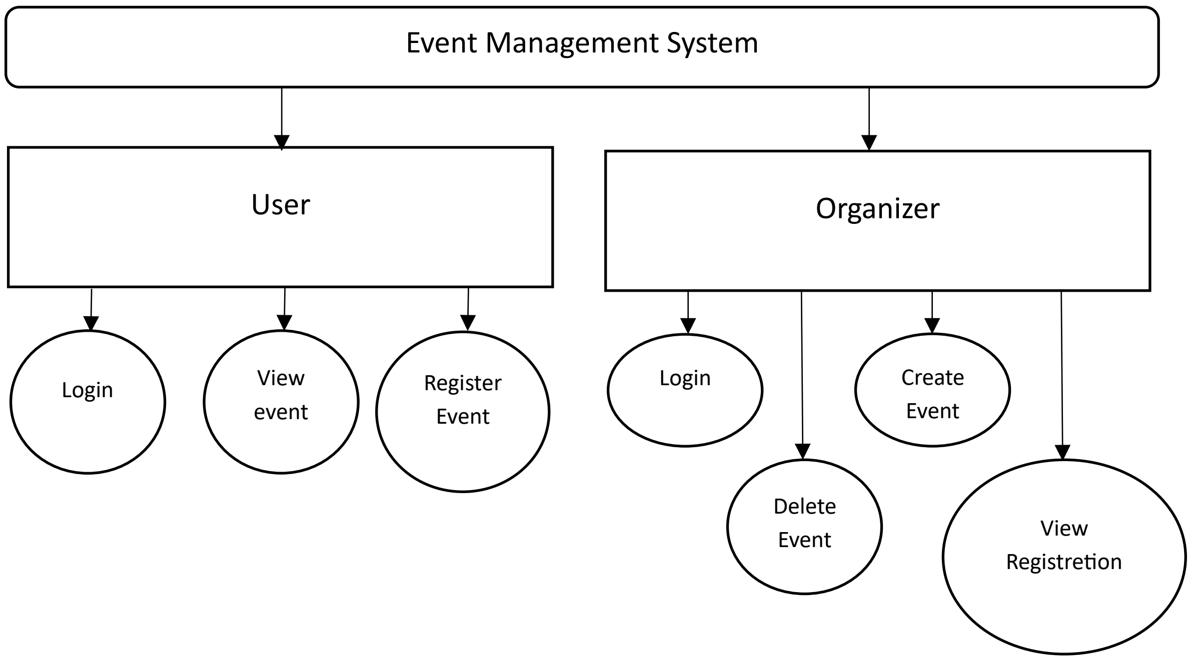
Both entities communicate with the Event Management System to perform different operations such as viewing events or managing events.



8.4 DFD Level 1 :

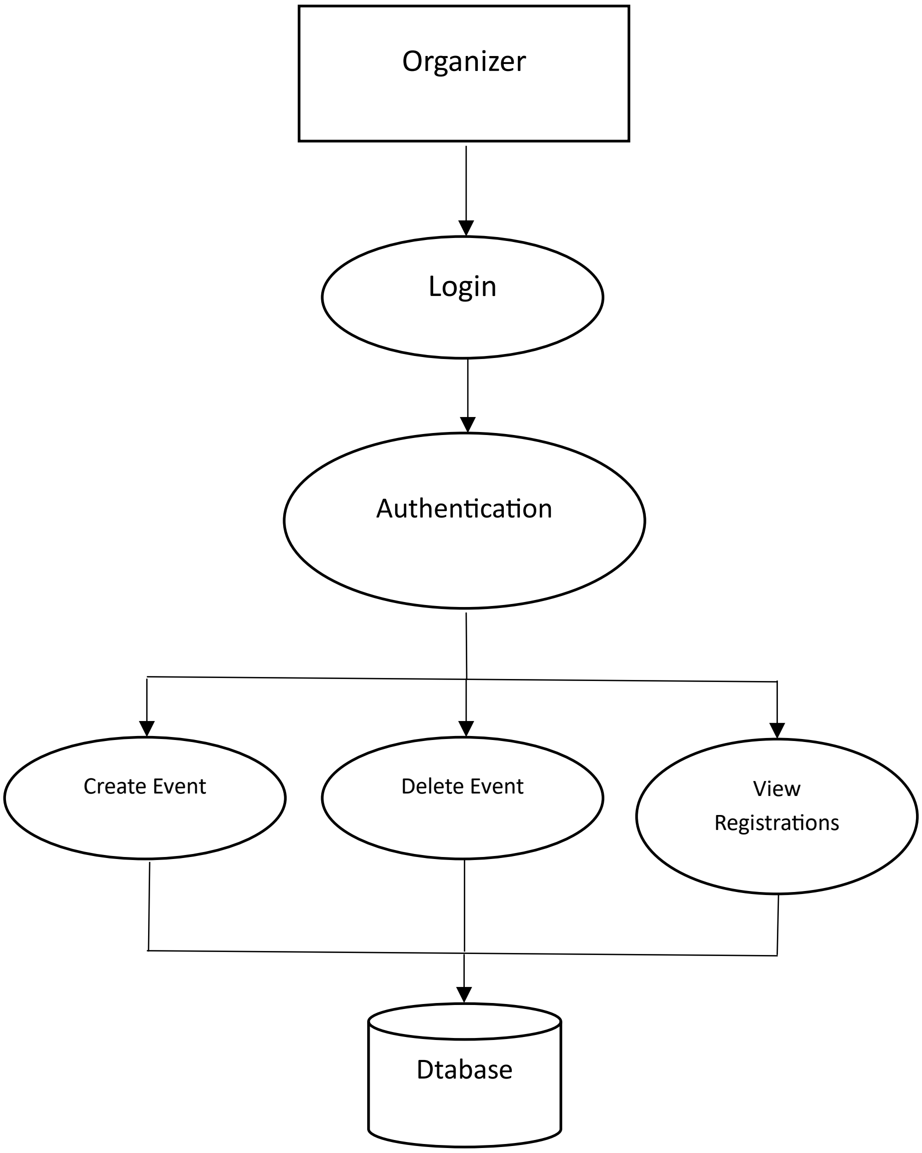
The Level 1 DFD provides more detailed information about the internal processes of the system.

- User operations include: Login , View Events , Register for Event
- Organizer operations include: Login , Create Event , Delete Event , View Registrations



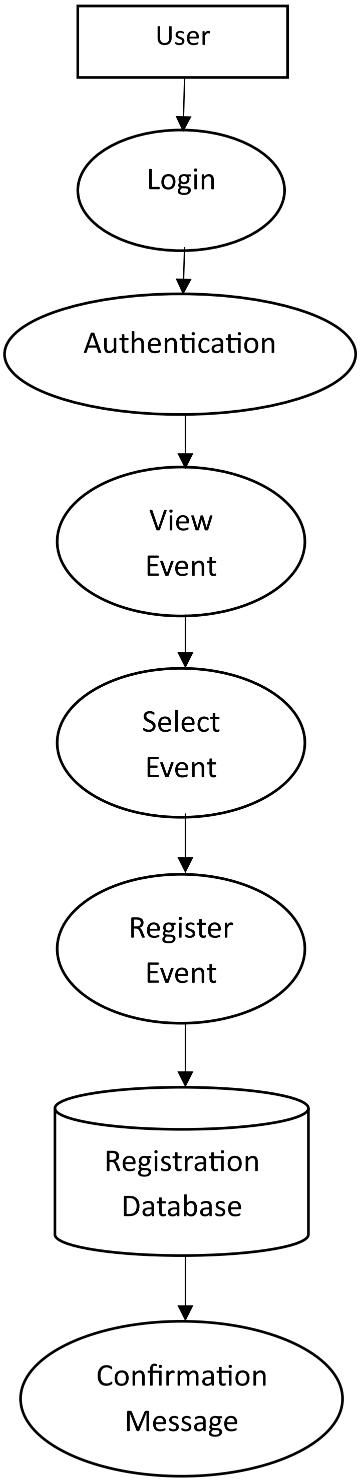
8.5 DFD Level 2 – Organizer :

Shows how the organizer logs in, creates events, deletes events, and views event registrations through the system database.



8.6 DFD Level 2 – User

Shows how the user logs in, views available events, selects an event, and registers for participation.

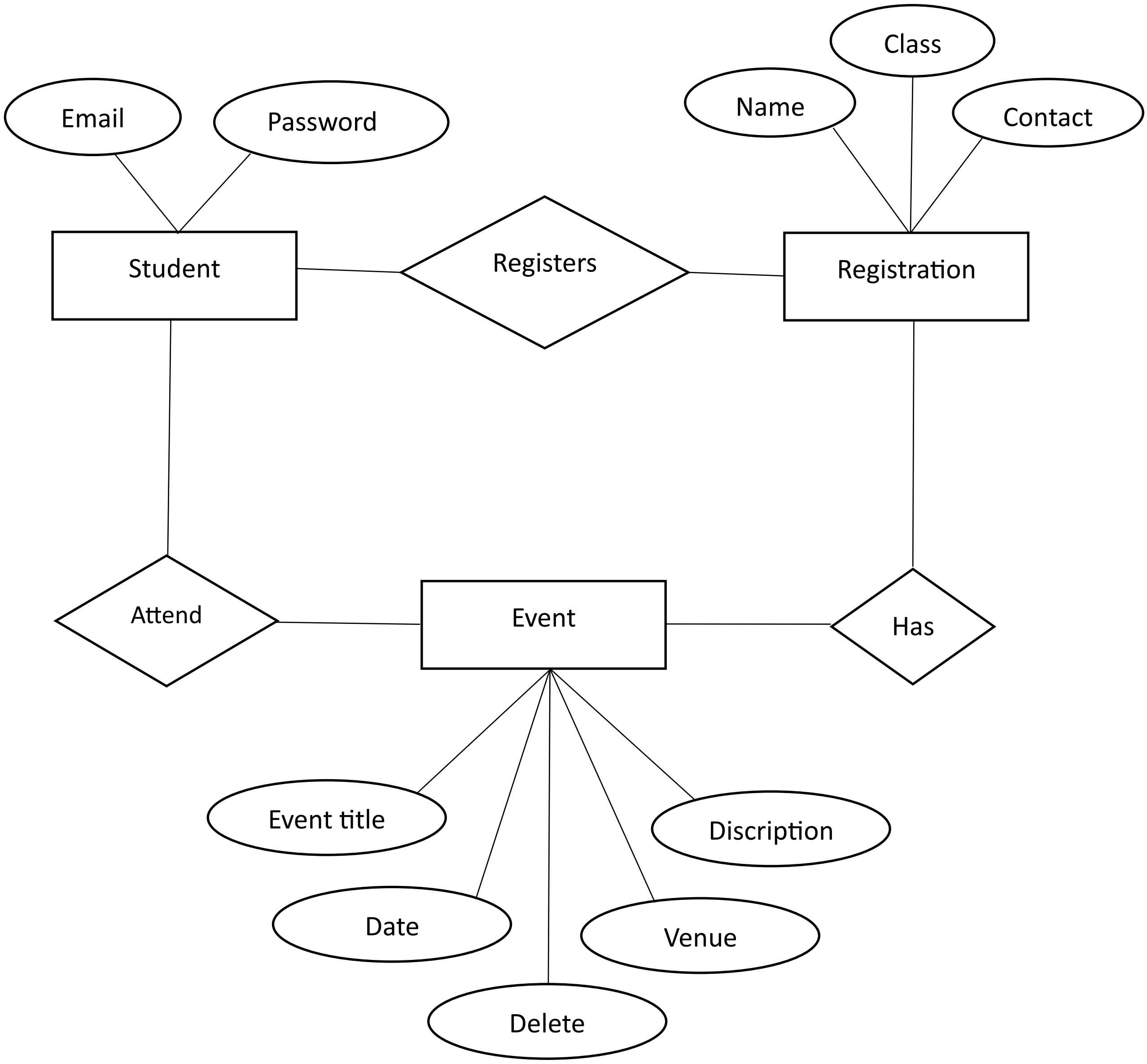


8.7 ER Diagram (Entity Relationship Diagram):

The ER diagram represents the relationship between different data entities in the system.

The main entities in the system are:

- User
- Event
- Registration



CHAPTER 9 : DATABASE DESIGN

9.1 Introduction:

Database design is an essential part of any information system because it defines how data is stored, organized, and accessed. A well-structured database ensures efficient data retrieval, consistency, and security.

In the College Event Management System, the database stores important information such as user accounts, event details, and event registrations.

9.2 Events:

The events table stores details about events created by organizers.

ADD DATA | UPDATE | DELETE | EXPORT DATA | EXPORT CODE | 25 | 1 - 2 of 2

events

	_id ObjectId	title String	date String	venue String	description String	or
1	ObjectId('69aec4084a6dde9…	"PROJECT PRESENTATION"	"2026-03-11"	"KHADSE COLLEGE ,MUKTAI N…	"All students who are in …	"o
2	ObjectId('69aec5d64a6dde9…	"TABLE TENNIS"	"2026-03-18"	"SGGKM SPORT CLUB "	"student who are interest…	"o

9.3 Event Registrations :

The event registration table stores information about users who register for events.

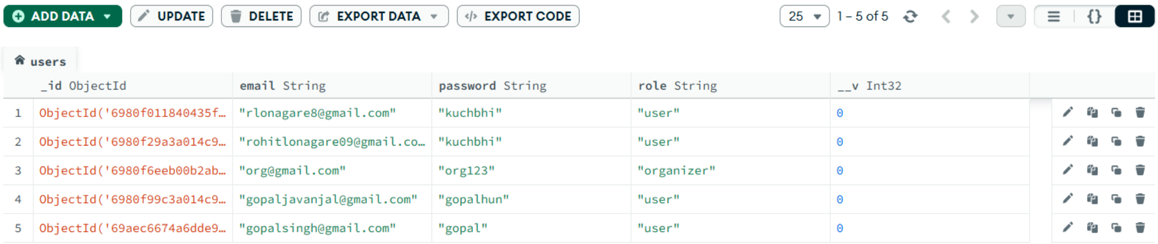
ADD DATA | UPDATE | DELETE | EXPORT DATA | EXPORT CODE | 25 | 1 - 8 of 8

eventregistrations

	_id ObjectId	eventId String	userName String	contact String	className String	_
1	ObjectId('69aec6e84a6dde9…	"69aec4084a6dde9e66e83e7a"	"Rohit R. Lonagare"	"xxxxxxxxxx"	"TY BCA"	0
2	ObjectId('69aec7344a6dde9…	"69aec4084a6dde9e66e83e7a"	"Shubhman Gill"	"xxxxxxxxxx"	"TY BCA"	0
3	ObjectId('69aec77c4a6dde9…	"69aec4084a6dde9e66e83e7a"	"Brean Larra"	"xxxxxxxxxx"	"TY BCA"	0
4	ObjectId('69aec7b04a6dde9…	"69aec4084a6dde9e66e83e7a"	"Kapil Dev"	"xxxxxxxxx"	"TY BCA"	0
5	ObjectId('69aec7f44a6dde9…	"69aec4084a6dde9e66e83e7a"	"Pat Cumins"	"xxxxxxxxxx"	"TY BCA"	0
6	ObjectId('69aec84a4a6dde9…	"69aec4084a6dde9e66e83e7a"	"Sachin Tendulkar"	"xxxxxxxxxx"	"TY BCA"	0
7	ObjectId('69aec86b4a6dde9…	"69aec4084a6dde9e66e83e7a"	"Thala"	"xxxxxxxxxx"	"TY BCA"	0
8	ObjectId('69aec8a34a6dde9…	"69aec4084a6dde9e66e83e7a"	"Bret Liee"	"xxxxxxxxxx"	"TY BCA"	0

9.4 User Registrations :

The user table stores information related to registered users in the system. This includes login credentials and user roles.

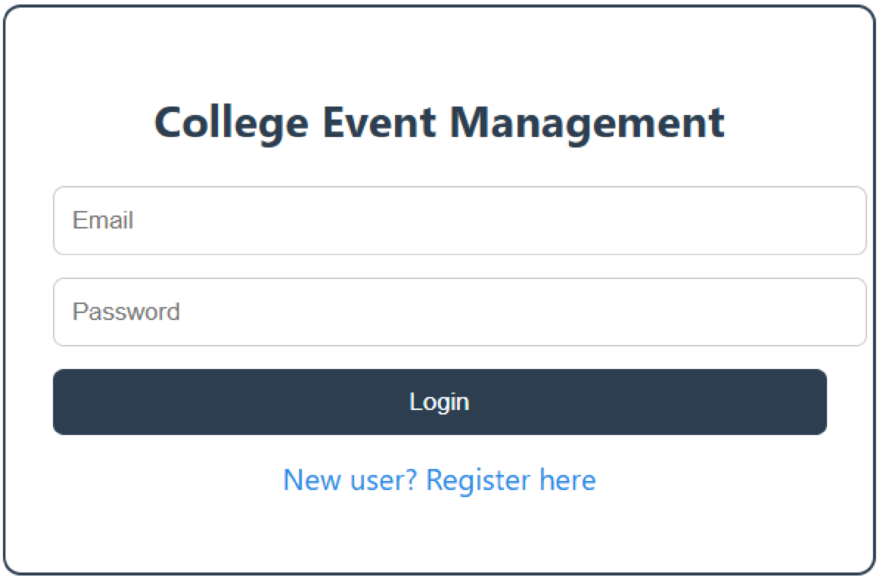


	_id ObjectId	email String	password String	role String	__v Int32
1	ObjectId('6980f011840435f…	"rlonagare8@gmail.com"	"kuchbhi"	"user"	0
2	ObjectId('6980f29a3a014c9…	"rohitlonagare09@gmail.co…	"kuchbhi"	"user"	0
3	ObjectId('6980f6eeb00b2ab…	"org@gmail.com"	"org123"	"organizer"	0
4	ObjectId('6980f99c3a014c9…	"gopaljavanjal@gmail.com"	"gopalhun"	"user"	0
5	ObjectId('69aec6674a6dde9…	"gopalsingh@gmail.com"	"gopal"	"user"	0

CHAPTER 10 : SNAPSHOTS

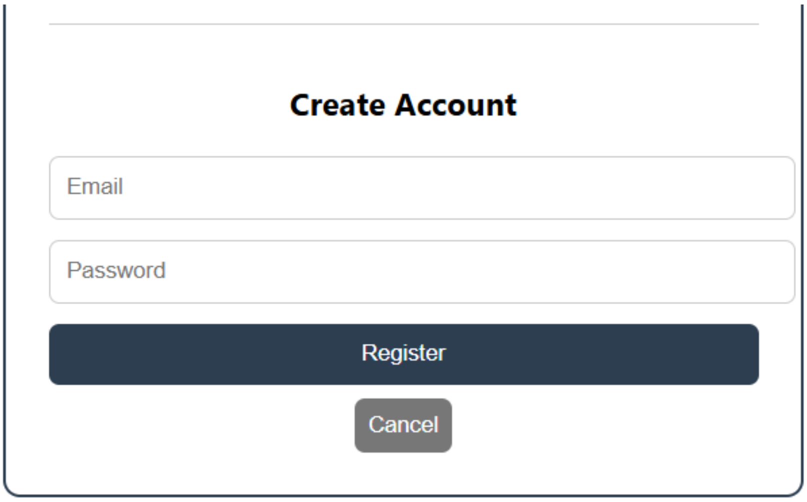
10.1 Login page :

This screen shows the login page of the Event Management System. Both users (students) and organizers (admins) can log in from the same interface by entering their registered email and password. After successful authentication, users are redirected to their respective dashboards where they can access system features.



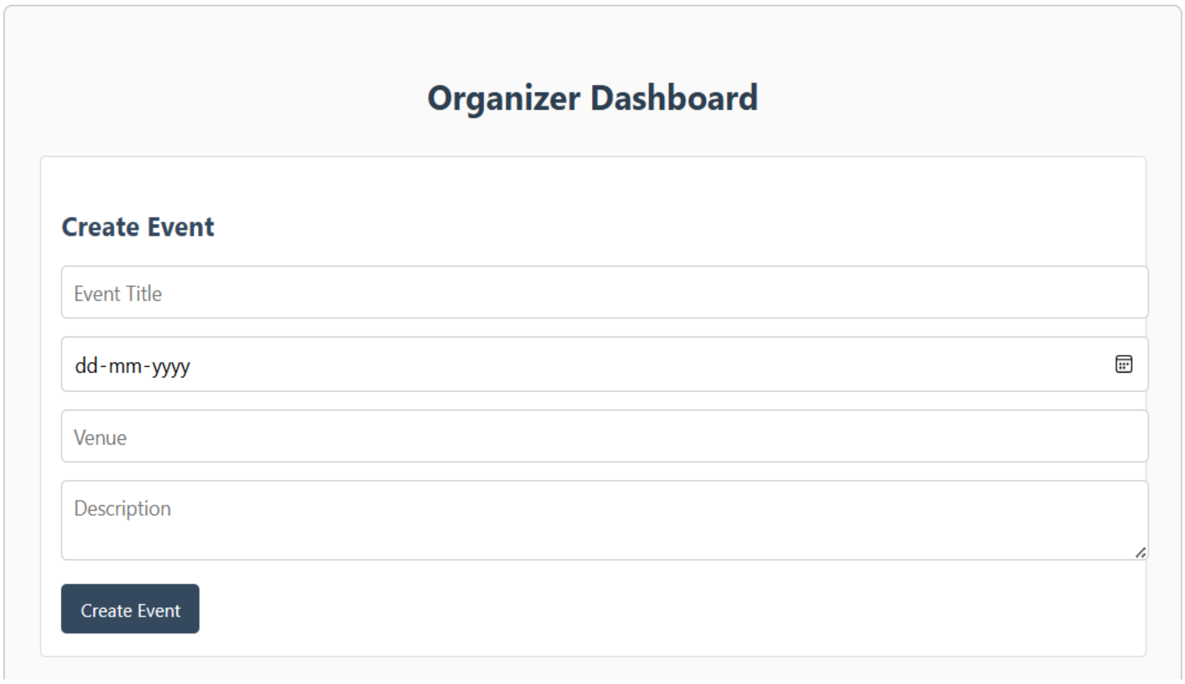
10.2 Registration page:

This screen displays the user registration page where new users can create an account by entering their details such as email, and password. After successful registration, the user can log in and access the Event Management System.



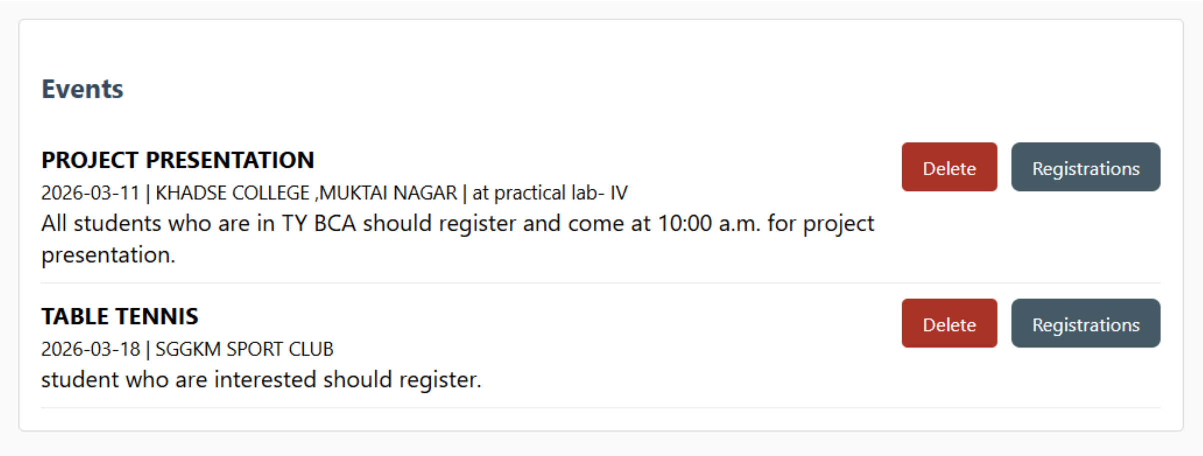
10.3 Event Creation :

This screen shows the organizer dashboard where the organizer can create and add new events to the system by entering event details such as title, date, location, and description.



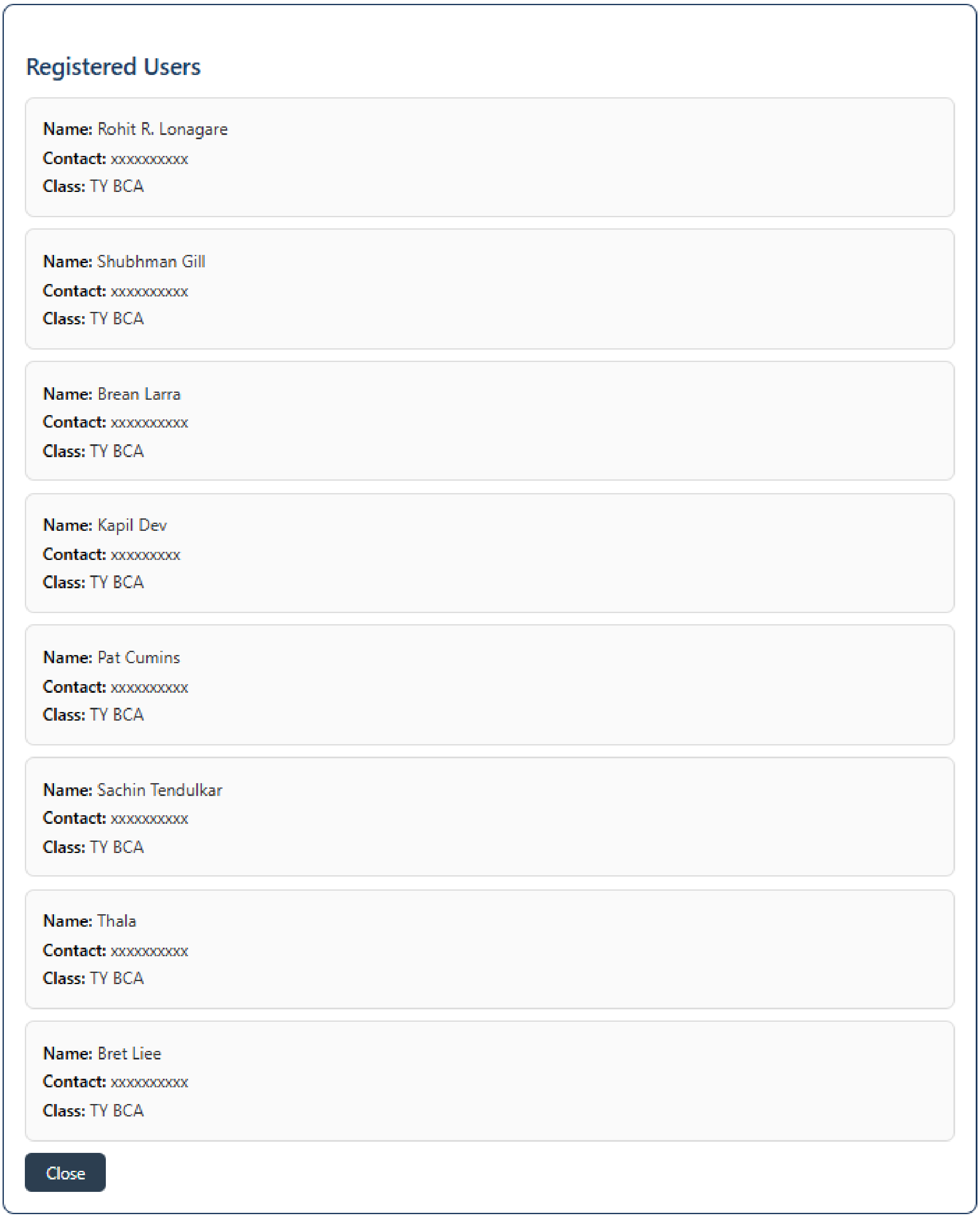
10.4 Event management :

This screen displays the list of events created by the organizer. The organizer can manage events by viewing event details, deleting events, and checking the list of registrations for each event.



10.5 View Registrations :

This screen displays the list of users who have registered for a particular event. The organizer can view the registration details of all participants who have successfully registered for the event.

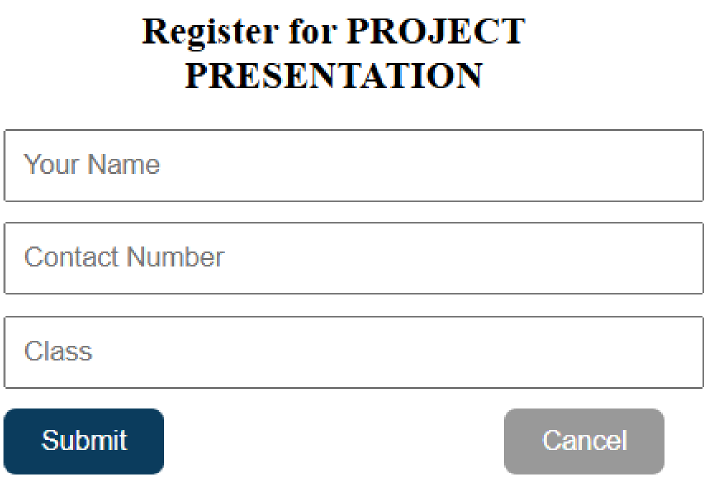


10.6 User Dashboard – Events List:

This screen displays the list of available events on the user dashboard. Users can view event details and register for a selected event by clicking the registration button.

10.7 Event Registration :

This screen shows the event registration form where users can register for an event by entering their details such as name, contact number, and class. The submitted information is stored in the system database.



CHAPTER 11 : SYSTEM TESTING

11.1 Introduction to System Testing

System testing is a crucial phase in the software development life cycle that ensures the developed system meets its specified requirements and functions correctly in a real-world environment. From a theoretical perspective, system testing validates the integration of various modules and confirms that the complete application behaves as expected under different scenarios.

In the College Event Management System, system testing plays a vital role because the application involves multiple user roles, dynamic data processing, and continuous interaction between frontend and backend components. Without systematic testing, errors in authentication, event handling, or data storage could compromise the reliability of the entire system.

11.2 Importance of System Testing

Theoretical studies in software engineering emphasize that system testing is essential for identifying defects that cannot be detected during unit or module testing. It focuses on verifying the system as a whole rather than individual components.

In this project, system testing ensures that:

- Users can register and log in successfully
- Organizers can create, view, and manage events
- Users can register for events
- Data is accurately stored and retrieved from the database

System testing helps confirm that the system performs consistently and reliably under normal usage conditions.

11.3 Objectives of System Testing

The primary objective of system testing is to validate the overall functionality and performance of the system. Theoretical frameworks define system testing as a process aimed at evaluating whether the system complies with functional and non-functional requirements.

For the College Event Management System, the objectives of system testing include:

- Verifying correct user authentication and authorization

- Ensuring accurate interaction between frontend and backend
- Confirming proper database operations
- Validating user workflows such as event creation and registration

These objectives guide the testing process and help ensure a robust final product.

11.4 Types of Testing Applied

System testing involves various testing techniques, each focusing on a specific aspect of the system. Theoretical literature categorizes testing into multiple types to ensure comprehensive coverage.

11.6.1 Functional Testing

Functional testing verifies whether the system performs its intended functions correctly. This form of testing focuses on input-output behavior without considering internal code structure.

In the College Event Management System, functional testing ensures that:

- Login credentials are validated correctly
- Event creation operations store correct data
- Registration processes associate users with events accurately

This testing confirms that each feature behaves according to its defined requirements.

11.6.2 Integration Testing

Integration testing examines the interaction between different system components. Theoretical models emphasize that even if individual modules function correctly, issues may arise when they are combined.

In this system, integration testing ensures seamless communication between:

- Angular frontend and Node.js backend
- Backend logic and MongoDB database

This testing validates that data flows correctly across system layers.

11.6.3 Validation Testing

Validation testing checks whether the system satisfies user expectations and business requirements. Theoretical approaches define validation as ensuring that the right system is built.

For this project, validation testing confirms that the system provides meaningful features such as:

- Easy event browsing for users
- Simple registration process
- Efficient event management for organizers

Validation testing ensures that the system is user-centric and practical.

11.5 Testing of Authentication System

Authentication testing is critical in systems that handle multiple user roles. From a theoretical standpoint, authentication testing ensures secure and controlled access to system resources.

In the College Event Management System, authentication testing verifies:

- Correct user login behavior
- Prevention of invalid login attempts
- Role-based access control

This testing ensures that only authorized users can access specific functionalities.

11.6 Testing of Event Management Module

The event management module forms the core functionality of the system. System testing ensures that event creation, display, and deletion operate correctly.

Theoretical testing principles emphasize checking data consistency and workflow continuity. In this system, testing validates that:

- Events created by organizers are visible to users
- Deleted events are removed from the system
- Event data remains consistent across sessions

This ensures smooth system operation and data reliability.

11.7 Testing of Event Registration Module

The event registration module allows users to participate in events. From a theoretical perspective, this module must be tested carefully to ensure accurate data mapping.

System testing ensures that:

- User registration data is saved correctly
- Multiple users can register for the same event
- Organizer can view registered participants

This testing verifies the integrity of user-event relationships within the system.

11.8 Database Testing

Database testing focuses on validating data storage, retrieval, and integrity. Theoretical studies highlight database testing as essential for maintaining system reliability.

In this project, database testing ensures that:

- User data is stored correctly
- Event data is consistent
- Registration records are accurate

Database testing helps prevent data loss and ensures correct system behavior.

11.9 Error Handling and Exception Testing

Exception testing evaluates how the system responds to unexpected inputs or failures. Theoretical models emphasize that robust systems must handle errors gracefully.

In the College Event Management System, error handling testing ensures:

- Informative error messages are displayed
- System does not crash due to invalid input
- Data remains unaffected during failures

This testing improves system stability and user experience.

11.10 Performance Considerations

Performance testing evaluates how the system behaves under different loads. Theoretical research highlights performance testing as essential for scalability.

Although this project is designed for academic use, system testing confirms that:

- Multiple users can access events simultaneously
- Database queries execute efficiently
- Page load times remain acceptable

This ensures readiness for future expansion.

11.11 Security Testing

Security testing evaluates system vulnerability to unauthorized access. Theoretical principles emphasize securing sensitive data and preventing misuse.

In this system, security testing ensures:

- Passwords are protected
- Role-based access is enforced
- Unauthorized actions are restricted

This testing safeguards system data and user privacy.

11.12 Test Results and Observations

System testing results indicate that the College Event Management System functions reliably under normal usage conditions. All major features operate as intended, and system workflows remain consistent.

Testing observations demonstrate effective integration between frontend, backend, and database components.

CHAPTER 12 : CONCLUSION

12.1 Overview of the Study

The development of the College Event Management System represents a practical application of theoretical concepts studied in software engineering, database management, and web technologies. This project demonstrates how abstract ideas such as system analysis, modular design, database normalization, and system testing can be transformed into a fully functional web-based application.

From a literature perspective, the project bridges the gap between academic theory and real-world system implementation. It reflects how structured methodologies and design principles contribute to the development of a reliable and efficient information system.

12.2 Achievement of Project Objectives

One of the primary goals of this project was to design and implement a system capable of managing college events efficiently. The literature on event management systems emphasizes automation, accuracy, and accessibility, all of which have been successfully addressed in this project.

The system achieves its objectives by providing a centralized platform for event creation, registration, and monitoring. Users can easily interact with events, while organizers can manage event-related activities in a controlled manner. These features demonstrate the successful realization of the project's initial objectives through structured design and implementation.

12.3 Application of Theoretical Concepts

This project extensively applies theoretical concepts from multiple domains of computer science. Software engineering principles such as modularity, separation of concerns, and lifecycle-based development are evident throughout the system architecture.

Database design theories, including normalization, entity relationship modeling, and data integrity, are reflected in the structured database collections. Similarly, web application concepts such as client-server architecture and role-based access control have been practically implemented.

The alignment between theory and practice reinforces the academic relevance of the project.

12.4 System Reliability and Performance

According to literature in software quality assurance, system reliability is a key indicator of project success. The College Event Management System demonstrates reliable performance under typical usage conditions.

Testing results confirm that the system handles user interactions, data processing, and event management tasks effectively. The stable interaction between frontend, backend, and database components supports consistent system behavior, validating the effectiveness of the applied design principles.

12.5 User-Centric Design Perspective

Modern literature emphasizes user experience as a critical factor in system acceptance. This project incorporates a user-centric approach by designing intuitive interfaces and clear workflows.

The system allows users to view events and register with minimal effort, while organizers are provided with efficient tools to manage events and monitor registrations. This design approach ensures that the system remains accessible and practical for its intended audience.

12.6 Security and Data Management Considerations

Security and data integrity are fundamental aspects highlighted in database and system design literature. The project addresses these concerns by implementing structured data storage, role-based access control, and controlled system interactions.

User credentials and event-related data are managed in a secure and organized manner, reducing the risk of unauthorized access or data inconsistency. This demonstrates the importance of integrating security considerations into system design from the initial stages.

12.7 Limitations of the System

Despite its strengths, the system has certain limitations, which are common in academic projects. Literature acknowledges that initial system versions often prioritize core functionality over advanced features.

The current system does not include features such as automated notifications, payment integration, or advanced analytics. However, these limitations do not reduce the educational value of the project; instead, they provide opportunities for further enhancement.

12.8 Scope for Future Enhancements

Future enhancement is an important component of system evaluation. Based on literature and current technological trends, the College Event Management System can be expanded in several ways.

Possible improvements include integration of notification services, enhanced security mechanisms, mobile application development, and cloud-based deployment. These enhancements would increase system scalability and usability, aligning it with modern enterprise-level applications.

12.9 Academic and Practical Significance

From an academic standpoint, this project serves as a comprehensive case study in applying theoretical knowledge to practical system development. It reinforces core concepts of software engineering and database management.

Practically, the system demonstrates how technology can streamline event management processes in educational institutions. This dual significance highlights the value of the project in both academic and real-world contexts.

CHAPTER 13 : REFERANCE

The following references were used during the development of this project.

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2. Angular Official Documentation.
   https://angular.io/docs
3. MongoDB Official Documentation.
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   https://www.tutorialspoint.com
6. Web development books and college lecture notes.